The Power of Letting Go: Scheduling, Rejection and Controlled Abandonment in Multi-Server Queues

Arsenii Shmelev, Noa Zychlinski
Faculty of Data and Decision Sciences,
Technion – Israel Institute of Technology, Haifa 3200003, Israel
ORCID: 0000-0002-5125-3089, noazy@technion.ac.il

Long waiting times in service and healthcare systems reduce customer satisfaction, increase abandonment rates, and harm provider profitability and reputation. To address these challenges, service providers explore various creative solutions and control strategies. We study a multi-server, multi-class scheduling problem in which the operator can either reject some arriving customers or initiate controlled abandonment to minimize total operational costs.

Using a fluid approximation, we develop an asymptotically optimal index-based policy for scheduling classes that simultaneously optimizes rejection and controlled abandonment decisions. We find that under optimal capacity allocation, each class may operate in a different queue regime based on its characteristics, following either an Erlang-B queue, where customers are rejected if immediate service is unavailable, or as an Erlang-A queue, where customers wait for service or abandon the queue.

When class-specific service-level (SL) requirements are included, the class indices are adjusted, impacting prioritization. Under the adjusted policy, strict priority is relaxed, allowing for partial capacity allocation for some classes. In such cases, the remaining capacity is allocated to a class with a lower index but a stricter SL requirement. Translating the fluid-based policy back into the original stochastic system results in a two-stage index rule. Simulation experiments confirm the effectiveness of our proposed policies compared to a commonly accepted benchmark policy.

Key words: scheduling queues, queues with rejection, service-level requirements, fluid models

1. Introduction

Long waiting times, a frequent issue in many service and healthcare systems, diminish customer satisfaction, increase the likelihood of customer abandonment, and can negatively impact both provider profitability and reputation (Jones and Peppiatt 1996, Silvester et al. 2004, Michael et al. 2013, Bleustein et al. 2014, Worthington 2004). To mitigate these effects, service providers can increase their service capacity or staffing levels (Harrison and Zeevi 2005, Whitt 2006), though this is not always feasible or cost-effective. Another option is to improve scheduling processes to prioritize certain customer classes over others (Atar et al. 2010, Huang et al. 2015). Alongside scheduling, in this paper we study the options to reject certain arriving customers or initiate controlled abandonment of waiting customers. The rationale behind these two strategies is to manage customer flow more effectively and maintain a high service level for those currently being served.

Specifically, these mechanisms allow the system operator to redirect certain customer classes either upon arrival or while they are waiting by dispatching them to a different service provider. In healthcare, for example, patients might be send to an alternative healthcare unit/provider. In call centers arriving customers or customer who have been waiting for an extended period may be removed from the system, effectively ending their waiting time without receiving the intended service. By redirecting or removing customers, the system can reduce congestion, optimize resource utilization, and improve overall service efficiency. These strategies also highlight the trade-offs between service availability and system efficiency, as some customer classes will receive a better service level, while others may receive worse service or none at all from the specific provider.

Several applications are relevant for our model, where rejecting arriving customers or initiating controlled abandonment is both reasonable and beneficial. In healthcare management, particularly in emergency departments or specialized clinics, rejecting or redirecting non-critical patients to other facilities ensures that critical cases receive immediate attention. In customer service call centers, rejecting or redirecting calls from lower-priority customers can prioritize high-priority calls, thereby maintaining service quality for the customers being served. In manufacturing, production lines may reject additional orders to prevent overloading and ensure timely delivery of existing commitments. In IT and cloud services, limiting access to non-critical users maintains operational integrity for essential services. Airlines frequently overbook flights and manage capacity by rejecting passengers with flexible travel plans, optimizing revenue and ensuring maximum seat utilization. Lastly, in the hospitality industry, rejecting last-minute bookings helps guarantee availability for higher-value customers or loyal guests, thereby enhancing customer satisfaction and optimizing resource allocation.

The paper's contribution can be summarized as follows:

- *Modeling.* We study a multi-server queue with multiple customer classes, relevant to various service applications, where each class has service-level (SL) requirements that must be satisfied. We explore how to optimally coordinate scheduling decisions under SL constraints using two operational controls: the option to reject arriving customers and the option to employ a controlled abandonment mechanism.
- *The $\mathcal{L}\mu$ Rule.* Using a fluid approximation, we derive an index-based rule – the $\mathcal{L}\mu$ rule – which balances each class's holding cost, service rate, and customer-initiated abandonment rate, as well as the rejection cost and operator-initiated abandonment cost. Additionally, we derive a closed-form

expression for the optimal rejection probability for each class. Beyond being easy to implement in practice, the suggested policy is also asymptotically optimal in the many-server heavy-traffic regime. Furthermore, we find that under the optimal policy and optimal capacity allocation, each class operates either like an Erlang-B queue, where customers who cannot be immediately served are rejected/blocked, or as an Erlang-A queue, where customers wait until they are served or abandon the queue. Simulation results demonstrate that the policy performs very well compared to a commonly accepted benchmark policy in various scenarios.

- *The Effect of SL Requirements.* When SL requirements for each class are introduced, the index-based $\mathcal{L}\mu$ rule is adapted into the $\mathcal{L}^{\mathrm{S}}\mu$ rule, ensuring that classes violating the constraint under the $\mathcal{L}\mu$ rule either receive higher prioritization or are more frequently rejected. Therefore, under the $\mathcal{L}^{\mathrm{S}}$ rule, the priority is no longer strict—it is possible that a class may not receive all the capacity it needs, while the remaining capacity is allocated to a class with a smaller $\mathcal{L}\mu$ index. The policy's translation to the original stochastic system involves a two-stage index policy, which ensures cost-effective scheduling while meeting all SL constraints.

The rest of the paper is organized as follows. Section 2 provides a brief review of the relevant literature. In Section 3, we introduce the stochastic model and the optimization problem. Section 4 presents the fluid model analysis, the optimality of the $\mathcal{L}\mu$ rule, and the translation of the policy back to the stochastic system. In Section 5, we integrate class-specific SL requirements and introduce the $\mathcal{L}^{\mathrm{S}}\mu$ rule and its translation back to the stochastic system. Finally, Section 6 concludes the paper and suggests several directions for future research. All proofs are provided in the appendix.

2. Literature Review

This paper is related to three primary areas of literature: the scheduling and resource allocation of queues with multiple customer classes, queueing systems with rejection, and service level with service level (SL) requirements. We will provide a brief review of the relevant literature in these areas.

Scheduling and resource allocation of queues with multiple customer classes. The scheduling of multiple customer classes within stochastic processing networks has been extensively studied. A foundational result in this area is the $c\mu$ rule, introduced by Cox and Smith (1961), which established the optimality of a simple index-based policy for a single-server queue with linear holding costs. Subsequent research has expanded on this rule, offering various generalizations. However, the optimality of these extensions is typically proven only in asymptotic regimes (e.g., Van Mieghem 1995, Mandelbaum and Stolyar 2004, Huang et al. 2015).

In multi-server systems, the dynamic scheduling of multiple classes with customer abandonment has been explored under different operational regimes. For example, Harrison and Zeevi (2004) and Atar et al. (2004) examined this problem under a critically loaded regime. Atar et al. (2010) further advanced this work by deriving the asymptotic optimality of the $c\mu/\theta$ rule for many-server queues with abandonment in the heavy traffic regime. More recent contributions, such as Long et al. (2020), have extended these scheduling rules to accommodate general queue length cost functions and varied customer patience distributions. Puha and Ward (2019) provided a comprehensive tutorial on scheduling policies for many-server queues with impatient customers in overloaded regimes. Other recent developments include the optimization of proactive service scheduling (Hu et al. 2022), and the management of new and returning patients in hybrid healthcare systems (Zychlinski 2024).

In this work, we complement this literature by studying the scheduling of multiple classes where the operator can either reject arriving customers or initiate abandonment of waiting patients.

Queueing Systems with Rejection. The concept of customer rejection in queueing systems due to finite capacity was first introduced by Erlang (1909), who studied blocking in telephone networks. This idea was further expanded in the context of loss networks by Palm (1943). In a comprehensive review, Stidham (1985) explored methods for controlling congestion in queueing systems by restricting arrivals, focusing on the distinction between socially optimal and individually optimal controls through both static and dynamic admission policies.

In the classical Erlang-B queueing system, customers are blocked or rejected if no capacity is available. However, rejection may also occur even when waiting space is available, when long wait times make timely service impossible. In this context, Perry and Asmussen (1995) analyzed an M/G/1 queue where customers may be partially or fully rejected based on their impatience and the system's state, deriving the steady-state distribution and exploring the busy period under various conditions. In an M/M/1 queue, Lewis (2001) investigated a queueing system where rejection occur either upon arrival or momentarily before service, demonstrating that optimal rejection policies exist and that the average reward increases when the second rejection recommendation is followed.

Queueing systems with rejection have broad applications, including in healthcare, telecommunications, manufacturing, and cloud computing. For instance, in healthcare, rejection is sometimes necessary when timely treatment cannot be guaranteed, as discussed by Allon et al. (2013), who examined ambulance diversion in emergency departments. In telephone call centers, Gans et al. (2003) reviewed staffing rules in an Erlang-B queueing systems, in different operational regimes.

More recently, Chydzinski (2023) studied a queueing system with probabilistic job rejection based on system occupancy, deriving formulas for transient and stationary throughput under various conditions, including different rejection probabilities, system loads, and inter-arrival distributions.

In a related vein, Legros (2020) explored a multi-server queueing system with a late rejection mechanism, similar to our suggested controlled abandonment. Their study identified an optimal time-based threshold policy for managing rejection, with the threshold determined by the system's state and cost metrics, such as waiting time percentiles.

Queueing Systems with Service Level Requirements. Service Level Agreements (SLAs) are formal commitments between a company and its customers, specifying the expected level of service, including response times and performance standards that the service provider must meet. These agreements are crucial for building trust, maintaining high service quality, and fostering long-term relationships with customers (Trienekens et al. 2004).

Most of the research regarding SL requirements has focused on staffing problems. Baron and Milner (2009), for example, introduced a period-based SLA framework for outsourced call centers, emphasizing short-term performance metrics such as rush hour delays and abandonment rates. These metrics were approximated and utilized to optimize staffing decisions, ultimately aiming to maximize profit.

More recently, Liu et al. (2024) addressed SL computation in a time-varying mixed-preemptive priorities queueing system, which features fluctuating arrival rates and multiple customer classes with varying priority rules. Their research provides exact and approximate methods for SL computation and applies these to an emergency department staffing problem, where service levels for different patient classes are treated as constraints. The study's computational tests, using real hospital data, demonstrate that the proposed staffing solutions effectively guarantee patients' SLs while reducing physicians' working hours.

In the current paper, we explore the optimal scheduling policy under class-specific SL constraints alongside determining the optimal rejection probability and controlled abandonment for each class.

3. The Model

We consider a Markovian N-server queuing model and I classes of customers, $\mathcal{I} = \{1, 2, \ldots, I\}$ who arrive to the system according to a time-homogeneous Poisson process with rate λ_i, $i \in \mathcal{I}$. Service and patience times of each class are exponential with rates μ_i and θ_i, $i \in \mathcal{I}$, respectively. Figure 1 illustrates our model.

Let $X_i(t)$ and $Q_i(t)$, $i \in \mathcal{I}$, denote the number of Class i customers in the system and in the queue, respectively, at time t, $t \geq 0$. A scheduling policy π determines the allocation of servers to customers. We consider Markovian non-anticipating policies; that is, server allocations are based on the current state $(X; Q)$ only. Under these scheduling policies, $\{(X(t); Q(t)) : t \geq 0\}$ is a Markov process. Let $Z_i(t)$ denote the number of servers occupied with Class i customers at time t.

In addition to the scheduling decision, we consider two operational controls. The first control includes the option to reject an arriving customer. The decision variable for this control is the rejection probability P_i for an arriving Class i customer. The rejection cost for each Class i customer is r_i, $i \in \mathcal{I}$. Once rejected, the customer leaves the system immediately without incurring further costs.

The second control we consider is the option to utilize a controlled abandonment mechanism, where the system operator can dispatch any waiting customer at any time. To model the controlled abandonment mechanism, we define an exponential random variable with rate $\hat{\theta}_i$, $i \in \mathcal{I}$, and allow for control over this rate. The associated cost of such controlled abandonment is denoted by $\hat{\alpha}_i$.

Therefore, there are two types of abandonment: endogenous (i.e., customer-initiated) abandonment stemming from customers' patience, and exogenous (i.e., operator-initiated) abandonment stemming from the operator's control.

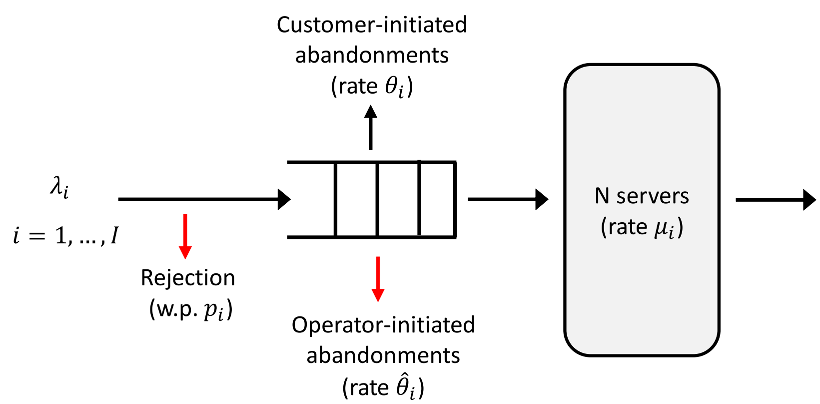

Figure 1 **An illustration of the model with I customer classes. The decision variables include the rejection probabilities and operator-initiated abandonment rate for each class, in addition to scheduling.**

Each class incurs a holding cost of h_i, $i \in \mathcal{I}$, per customer per unit of time. We incur an abandonment cost α_i, $i \in \mathcal{I}$ for each Class i customer who abandons while waiting in the queue. Let $R_i(t)$ and $\Gamma_i(t)$ represent the cumulative number of Class i customer- and operator-initiated abandonment, respectively, by time t. Additionally, define $A_i^P(t)$ as the cumulative number of rejected Class i customers up to time t.

The aggregated cost up to time T is, therefore,

$$\mathbb{E}\left[\int_0^T \left(\sum_{i\in\mathcal{I}} h_i Q_i(t)\right) \mathrm{d}t + \sum_{i\in\mathcal{I}} \left[r_i A_i^P(T) + \alpha_i R_i(T) + \hat{\alpha}_i \Gamma_i(T)\right]\right],$$

where under the Markovian modeling assumption,

$$\mathbb{E}\left[R_i(T)\right] = \theta_i \mathbb{E}\left[\int_0^T Q_i(t)\mathrm{d}t\right], \quad \mathbb{E}\left[\Gamma_i(T)\right] = \hat{\theta}_i \mathbb{E}\left[\int_0^T Q_i(t)\mathrm{d}t\right].$$

We, therefore, get that

$$\mathbb{E}\left[\int_0^T \left(\sum_{i\in\mathcal{I}} \left[(h_i+\alpha_i\theta_i)\,Q_i(t)+\hat{\alpha}_i\hat{\theta}_i Q_i(t)\right]\right)\mathrm{d}t+\sum_{i\in\mathcal{I}} r_i A_i^P(T)\right],$$

For simplicity of notation we introduce the "generalized" holding costs $c_i = h_i + \alpha_i\theta_i,\ i\in\mathcal{I}$. Our goal is to find a scheduling policy π that minimizes the total expected long-run average loss, specifically:

$$\begin{aligned}
&\min_{\pi\in\Omega,P,\hat{\theta}}\ \limsup_{T\to\infty}\frac{1}{T}\mathbb{E}\left[\int_0^T \left(\sum_{i\in\mathcal{I}}\left[c_i Q_i(t)+\hat{\alpha}_i\hat{\theta}_i Q_i(t)\right]\right)\mathrm{d}t+\sum_{i\in\mathcal{I}} r_i A_i^P(T)\right]\\
&\text{s.t.}\quad \sum_{i\in\mathcal{I}} Z_i(t)\le N;\\
&\qquad\ 0\le Z_i(t)\le X_i(t),\ i\in\mathcal{I},\ \forall t\ge 0,
\end{aligned}\tag{1}$$

where Ω denotes the set of admissible controls.

Problem (1) is a Markov Decision Process (MDP). However, the curse of dimensionality – stemming from the need to manage a large (or potentially infinite) state and policy space – complicates the identification and characterization of the optimal scheduling policy (Papadimitriou and Tsitsiklis 1999).

To gain structural insights into the optimal scheduling policy and capacity allocation, we employ a deterministic fluid model. Fluid models are valued for their ability to approximate the first-order average behavior of stochastic systems, making them useful in various service operations management applications (Whitt 2002, Zychlinski 2022). These models are typically derived as asymptotic limits through the Functional Law of Large Numbers. In this paper, we adopt the many-server heavy traffic framework, where both the arrival rate and the number of servers scale proportionally.

4. The Fluid Model

In the fluid model, deterministic continuous rates replace stochastic processes. We use lowercase $\bar{x}_i, \bar{q}_i$ and $\bar{z}_i$, $i\in\mathcal{I}$, to denote the *average* fluid content in the system, the queue length, and the *fraction* of the service capacity assigned to Class i, respectively. The fraction of Class i customers who are rejected is p_i.

The above quantities satisfy the following relations for each $i\in\mathcal{I}$:

$$\begin{cases} \lambda_i(1-p_i)=\mu_i\bar{z}_i+\left(\theta_i+\hat{\theta}_i\right)\bar{q}_i;\\ \bar{q}_i,\bar{z}_i\ge 0. \end{cases}$$

We define the fluid analogue problem to the MDP in (1) as

$$\begin{aligned}
&\min_{\bar{q},\bar{z},p,\hat{\theta}} \sum_{i\in\mathcal{I}} \left[\left(c_i+\hat{\alpha}_i\hat{\theta}_i\right)\bar{q}_i + r_i\lambda_i p_i\right]\\
&\text{s.t.} \quad \lambda_i\left(1-p_i\right) = \mu_i\bar{z}_i + \left(\theta_i+\hat{\theta}_i\right)\bar{q}_i,\\
&\qquad \sum_{i\in\mathcal{I}} \bar{z}_i \le 1, \quad \bar{q}_i, \bar{z}_i \ge 0, \quad i\in\mathcal{I}.
\end{aligned}$$

Noting that the first constraint is equivalent to

$$\bar{q}_i = \frac{\lambda_i(1-p_i)-\mu_i\bar{z}_i}{\theta_i+\hat{\theta}_i}, \tag{2}$$

and substituting it into the objective function yields the following

$$\min_{\bar{q},\bar{z},p,\hat{\theta}} \sum_{j=1}^{\mathcal{I}} \left[\frac{\lambda_i(1-p_i)\left(c_i+\hat{\alpha}_i\hat{\theta}_i\right)}{\theta_i+\hat{\theta}_i} - \frac{\left(c_i+\hat{\alpha}_i\hat{\theta}_i\right)\mu_i}{\theta_i+\hat{\theta}_i}\bar{z}_i + r_i\lambda_i p_i\right],$$

which can be rewritten as

$$\min_{\bar{q},\bar{z},p,\hat{\theta}} \sum_{i\in\mathcal{I}} \left[\frac{\lambda_i(c_i+\hat{\alpha}_i\hat{\theta}_i)}{\theta_i+\hat{\theta}_i} - \lambda_i\left(\frac{(c_i+\hat{\alpha}_i\hat{\theta}_i)}{\theta_i+\hat{\theta}_i} - r_i\right)p_i - \frac{(c_i+\hat{\alpha}_i\hat{\theta}_i)\mu_i}{\theta_i+\hat{\theta}_i}\bar{z}_i\right].$$

Defining

$$\hat{c}_i := \frac{c_i+\hat{\alpha}_i\hat{\theta}_i}{\theta_i+\hat{\theta}_i}, \tag{3}$$

we get the following optimization problem:

$$\begin{aligned}
&\max_{\bar{z},p,\hat{\theta}} \sum_{i=1}^{\mathcal{I}} \left[\hat{c}_i\mu_i\bar{z}_i + \lambda_i\left(\left(\hat{c}_i-r_i\right)p_i-\hat{c}_i\right)\right]\\
&\text{s.t.} \quad \sum_{i\in\mathcal{I}} \bar{z}_i \le 1,\\
&\qquad 0\le \bar{z}_i \le \lambda_i\left(1-p_i\right)/\mu_i, \quad p_i\in[0,1].
\end{aligned} \tag{4}$$

4.1. Optimal Fluid Solution

We start by defining the following index for each customer class:

$$\mathcal{L}_i := \left(r_i \wedge \inf_{\hat{\theta}_i} \hat{c}_i\right), \quad i\in\mathcal{I}, \tag{5}$$

where $x\wedge y = \min(x,y)$ and $\hat{c}_i$ is as defined in (3).

Next, we move to the following definition and outline some useful properties of the optimal solution.

Definition 1 (Rejectable and Non-Rejectable Classes) *Class i is considered rejectable if $\mathcal{L}_i = r_i$, and non-rejectable, otherwise.*

Proposition 1 *Under the optimal solution of (4):*

- *For any class $i \in \mathcal{I}$, the following holds:*

$$\hat{c}_i^* = \inf_{\hat{\theta}_i} \hat{c}_i = \begin{cases} c_i/\theta_i, & \text{if } c_i/\theta_i \leq \hat{\alpha}_i, \\ \hat{\alpha}_i, & \text{otherwise.} \end{cases}$$

where $\hat{\theta}_i^* = 0$ or tends to ∞, respectively.

- *For any rejectable class $i \in \mathcal{I}$, the optimal queue length is $\bar{q}_i^* = 0$.*
- *For any non-rejectable class $i \in \mathcal{I}$ optimal rejection probability $p_i^* = 0$.*

Note that, according to Proposition 1, the optimal controlled abandonment rate, $\hat{\theta}$, can take only extreme values – either 0 or ∞. When $\hat{\theta} = 0$, the abandonment mechanism is effectively inactive, resulting in no operator-initiated abandonments. Conversely, as $\hat{\theta} \to \infty$, the system immediately rejects all incoming customers. Hence, the rejection mechanism inherently encompasses the controlled abandonment process. This is not the case, however, when SL requirements are introduced, as demonstrated in Section 5.

Next, we define the $\mathcal{L}\mu$ index rule as follows:

Definition 2 (the $\mathcal{L}\mu$ rule) *Assign priority and service capacity to Class i, $i \in \mathcal{I}$, having the higher $\mathcal{L}_i\mu_i$ index.*

Theorem 1 establishes the optimality of the $\mathcal{L}\mu$ rule as the solution to (4).

Theorem 1 (optimality of the $\mathcal{L}\mu$ rule) *For the optimization problem (4), the $\mathcal{L}\mu$ rule is optimal. The optimal rejection probability is*

$$p_i^* = 1_{\left\{\inf_{\hat{\theta}_i} \hat{c}_i \geq r_i\right\}} \left(1 - \frac{\mu_i}{\lambda_i} \bar{z}_i\right) \tag{6}$$

where $1_{\{x\}}$ is an indicator function that equals 1 when the condition x is true, and 0 otherwise.

Notice that when only the rejection control is available, the $\mathcal{L}$-index and the optimal rejection probability reduce to:

$$\mathcal{L}_i := \left(r_i \wedge \frac{c_i}{\theta_i}\right), \text{ and } p_i^* = 1_{\left\{\frac{c_i}{\theta_i} \geq r_i\right\}} \left(1 - \frac{\mu_i}{\lambda_i} \bar{z}_i\right).$$

If we make both rejection and control abandonment ineffective by setting r and $\hat{\alpha}$ higher then c/θ, Theorem 1 reduces to the well-known $c\mu/\theta$ rule (Atar et al. 2010).

The implications of the $\mathcal{L}\mu$ rule differ for rejectable and non-rejectable classes. For rejectable classes, all customers who cannot be immediately served are rejected. In contrast, for non-rejectable classes, neither customer is rejected. Abandonment, in this case, can be either endogenous (stemming from customers' patience) or exogenous (stemming from the operator's control). This distinction arises from Proposition 1: for non-rejectable classes, $\hat{\theta}$ is either 0 (indicating no controlled abandonment) or tends to ∞ (indicating instant exogenous abandonment).

When all $\mathcal{L}\mu$ indices are distinct, at most one class—the one that cannot be fully served—can have an optimal rejection probability strictly between 0 and 1. This is due to the optimal solution dictating that classes with a higher index are fully served, resulting in $p_i = 0$, while classes with a lower index are not served, leading to either immediate rejection ($p_i = 1$) or (exogenous/endogenous) abandonment ($p_i = 0$).

These results reveal an interesting insight. Recall that $\mathcal{L}_i = \left(r_i \wedge \inf_{\hat{\theta}_i} \hat{c}_i\right)$. If $\mathcal{L}_i = r_i$, then under optimal control, the system for Class i customers operates like an Erlang-B queue, where customers who cannot be immediately served are rejected/blocked. If, however, $\mathcal{L}_i = \inf_{\hat{\theta}_i} \hat{c}_i$, then the system for Class i operates as an Erlang-A queue, where customers wait until they are served or abandon the queue. In other words, each class in our system might operate according to a different queue regime, depending on its characteristics.

4.2. A Numerical Example

To illustrate the $\mathcal{L}\mu$ policy, we consider a three-class system with five servers. Figure 2 displays the average number of each customer class in the system and the average server allocation to each class as a function of Class 1's rejection cost, r_1. There are two priority switches, occurring at $r_1 = 10$ and $r_1 = 20$, resulting in three distinct regions. This arises because $\mathcal{L}_3 = c_3/\theta_3 = 10$, $\mathcal{L}_2 = c_2/\theta_2 = 20$, and $\mu_1 = \mu_2 = \mu_3 = 1$.

In the rightmost region, Class 1's rejection cost is sufficiently high, giving it the highest priority. As a result, no rejections occur, and Class 1 is fully served. In the middle region, priority switches between Class 1 and Class 2, with most servers allocated to Class 2. Here, 75% of arriving Class 1 customers are rejected. In the leftmost region, where the rejection cost is very low, priority switches between Class 1 and Class 3, leading to the complete rejection of Class 1 customers.

4.3. The Asymptotic Optimality of the $\mathcal{L}\mu$ Rule

In this section, we prove that the $\mathcal{L}\mu$ rule is asymptotically optimal in the heavy-traffic many-server regime. Given that we have established how optimal controlled abandonment can be effectively implemented via the rejection mechanism, the proof focuses solely on the rejection mechanism.

We consider a sequence of systems indexed by the number of servers, so that in the n-th system $X_i^n(t)$, $Q_i^n(t)$ and $Z_i^n(t)$, $i \in \mathcal{I}$ denote the respective total number of customers, queue length and number of allocated servers for Class i, such that $\forall t \geq 0$:

$$\sum_{i \in \mathcal{I}} Z_i^n(t) \leq n;$$

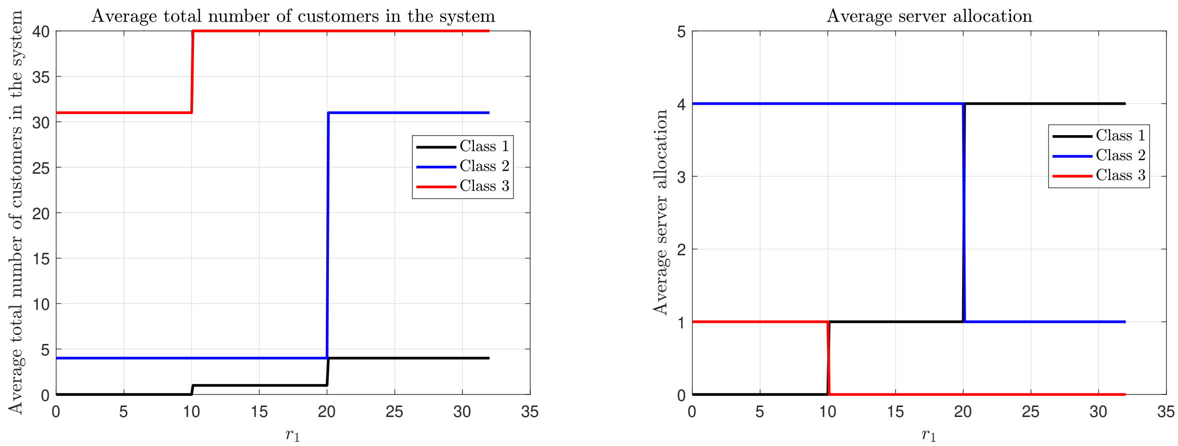

Figure 2 **Average number of customers in the system and server allocation for each class under the $\mathcal{L}\mu$ rule. The parameters are:** $N=5$, $\lambda=(4,4,4)$, $\mu=(1,1,1)$, $\theta=(0.1,0.1,0.1)$, $h=(2.8,1.8,0.8)$, $r=(r_1,30,30)$, $\alpha=(2,2,2)$, $\hat{\alpha}=(50,50,50)$.

$$X_i^n(t)-Z_i^n(t)=Q_i^n(t)\geq 0, \quad i\in\mathcal{I}.$$

The arrival process for each Class i, $i\in\mathcal{I}$, denoted by A_i^n, is modeled as a Poisson process with rate λ_i^n. The probability of rejecting an incoming Class i patient is P_i^n. We define the rejection and non-rejection processes as $A_i^{(P)n}$ and $A_i^{(1-P)n}$, respectively. By the properties of Poisson processes, both $A_i^{(P)n}$ and $A_i^{(1-P)n}$ are Poisson processes with rates $\lambda_i^n P_i^n$ and $\lambda_i^n(1-P_i^n)$, respectively.

Let $D_i^n(t)$ and $R_i^n(t)$ denote the number of Class i service completions and abandonments by time t, respectively, corresponding to Poisson processes $\tilde{D}_i^n$ and $\tilde{R}_i^n$ with rates $\mu_i^n>0$ and $\theta_i^n>0$, respectively. The above processes satisfy the following relation:

$$X_i^n(t)=X_i^n(0)+A_i^{(1-P)n}(t)-D_i^n(t)-R_i^n(t), \quad i\in\mathcal{I},\ t\geq 0. \tag{7}$$

Let π^n denote a policy for the n-th system, and Π^n denote the collection of all feasible policies for n-the system.

We define the normalized average cost as follows:

$$C_{n,T}(\pi^n):=\frac{1}{nT}\mathbb{E}\left[\int_0^T c\,Q^n(t)\,dt\ +\ rA^{(P)n}(T)\right], \tag{8}$$

where

$$V_{n,T}=\inf_{\pi^n\in\Pi^n} C_{n,T}(\pi^n).$$

Here, $c\,Q^n(t)=\sum_{i\in\mathcal{I}} c_i Q_i^n(t)$, and $rA^{(P)n}(T)=\sum_{i\in\mathcal{I}} r_i A_i^{(P)n}(T)$.

Lastly, we denote by q^*, z^* and p^* the optimal queue length, fraction of server capacity and rejection probability in the fluid model, which satisfy $x^*=q^*+z^*$. The optimal value function is, therefore,

$$V^*:=cq^*+r\lambda p^*.$$

We make the following assumptions:

Assumption 1 *(i) There exist positive constants* λ_i, μ_i, θ_i, $i \in \mathcal{I}$, *such that,* $\lambda_i^n/n \to \lambda_i$, $\mu_i^n \to \mu_i$, $\theta_i^n \to \theta_i$, *as* $n \to \infty$. *(ii) The rejection probabilities we consider satisfy* $P_i^n \to p_i$, $i \in \mathcal{I}$ *as* $n \to \infty$, *where* p_i *are non-negative constants. (iii) The random variables* $n^{-1}\mathbf{X}^n(0)$ *are uniformly bounded by a constant* M.

Assumption 1(ii) ensures that the rejection probabilities exhibit a consistent pattern relative to the system's size, thus preventing irregularities such as cyclic behavior.

Proposition 2 establishes that the $\mathcal{L}\mu$ rule is asymptotically optimal.

Proposition 2 (Asymptotic optimality) *Under Assumption 1,*

(a) $v := \liminf_{T\to\infty} \liminf_{n\to\infty} V_{n,T} \geq V^*$;

(b) $v_p := \limsup_{T\to\infty} \limsup_{n\to\infty} C_{n,T}(\pi^n_{\mathcal{L}\mu}) = V^*$, *where* $\pi^n_{\mathcal{L}\mu}$ *is the* $\mathcal{L}\mu$ *scheduling policy.*

4.4. Translation Back to the Stochastic System

The $\mathcal{L}\mu$ policy is derived through a fluid approximation. Our objective, however, is to implement this policy and evaluate its performance in the original stochastic system. Due to the ability to reject customers or utilize control abandonment, translating the fluid policy back to the stochastic system is not straightforward. Therefore, this section has two main objectives: First, to develop an effective and implementable translation of the policy, and second, to compare the performance of our proposed $\mathcal{L}\mu$ policy to a benchmark policy – the standard $c\mu/\theta$ policy.

Customer scheduling is managed through a strict priority rule, and while there are multiple ways to handle rejection, we focus on the one that most accurately reflects the fluid optimal policy. The optimal rejection rate for each class is derived based on the availability of immediate service. Specifically, an arriving Class i customer with $p_i^* > 0$ enters the system only if a server is available; otherwise, the customer is rejected. Therefore, this is how we implement the rejections in the simulation.

To evaluate the effectiveness of this policy, we utilize a stochastic simulation model that we developed. Figure 3 compares the fluid solution to the $\mathcal{L}\mu$ rule and the $c\mu/\theta$ rule in a simulation incorporating three customer classes. The figure illustrates the long-run average cost for different values of Class 1's rejection costs, r_1. The parameters we use are the same as in Section 2 and Figure 2.

We observe that the $\mathcal{L}\mu$ rule outperforms the $c\mu/\theta$ rule as long as the rejection cost is not excessively high. When the rejection cost is high, the performance of both rules converges, as no rejections occur. Notably, these differences persist regardless of system size. Additionally, the $\mathcal{L}\mu$ rule proves effective, performing very close to the optimal fluid solution.

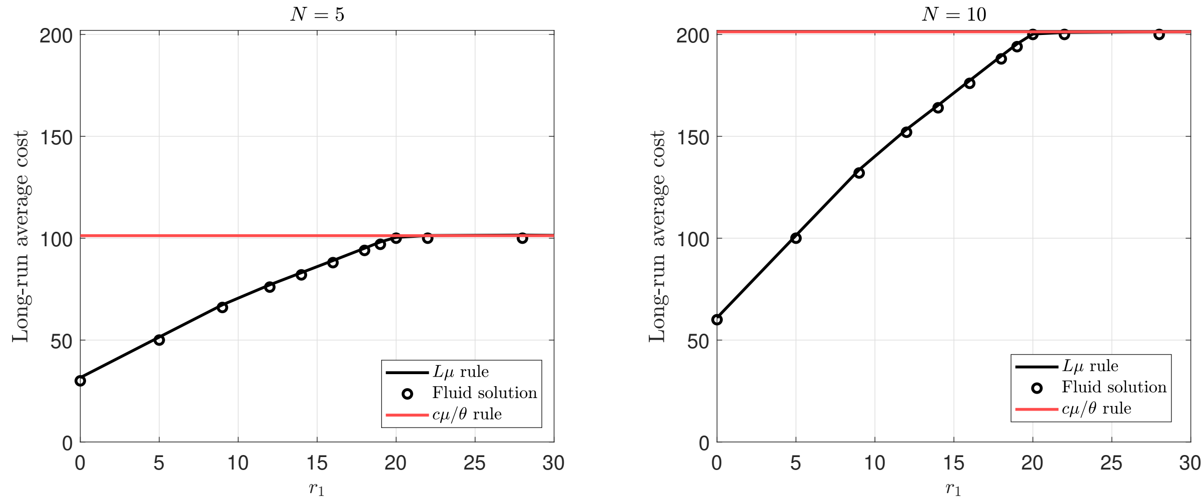

Figure 3 **Cost comparison between the $\mathcal{L}\mu$ implementation in the stochastic system, the fluid model, and the $c\mu/\theta$ rule for varying rejection costs. The parameters are:** $\mu = (1,1,1)$, $\theta = (0.1, 0.1, 0.1)$, $h = (2.8, 1.8, 0.8)$, $r = (r_1, 30, 30)$, $\alpha = (2,2,2)$, $\hat{\alpha} = (50, 50, 50)$. **In the left plot,** $N = 5$, $\lambda = (4,4,4)$**; in the right plot** $N = 10$, $\lambda = (8,8,8)$.

5. Integrating Class-Specific Service Level (SL) Requirements

In many service systems, maintaining a high standard of service quality is essential. To achieve this, SL requirements are often established for each customer class. The SL requirements considered in this paper impose an upper limit on the average waiting time that customers in each class can experience before receiving service. By setting these SL thresholds, service providers can ensure timely service and prevent excessive waiting times. In this section, we incorporate class-specific SL requirements into the scheduling problem, accounting for both rejection and controlled abandonment.

Let $\bar{w}_i$ denote average waiting time of Class i customers, and $\tau_i \geq 0$, $i \in \mathcal{I}$ denote the maximal average waiting time allowed for Class i customers. We, therefore, require that

$$\bar{w}_i \leq \tau_i, \quad i \in \mathcal{I}.$$

Applying Little's law, we get the following constraint:

$$\bar{w}_i = \frac{\bar{q}_i}{\lambda_i (1 - p_i)} \leq \tau_i.$$

Then, plugging in the steady-state equation in (2), yields the following equivalent

$$\frac{\lambda_i(1-p_i) - \mu_i \bar{z}_i}{\lambda_i (1-p_i) \left(\theta_i + \hat{\theta}_i\right)} \leq \tau_i,$$

which can be rewritten as

$$\bar{z}_i \geq \frac{\lambda_i}{\mu_i}(1-p_i)\left(1-(\theta_i+\hat{\theta}_i)\tau_i\right). \tag{9}$$

Trivially, if the solution to the problem without SL constraints already satisfies the SL constraints, then this solution remains optimal. Note also that per Proposition 1, in case $\mathcal{L}_i = r_i$ or $\mathcal{L}_i = \hat{\alpha}_i$, we have $\bar{q}_i = 0$, so the constraint is satisfied. Therefore, only when $\mathcal{L} = c_i/\theta_i$ (implying that optimally $\hat{\theta}_i = 0$ and $p_i = 0$), the SL constraint may be violated. We, therefore, introduce a specific subset of customer classes – the constraint-breaching Classes.

Definition 3 (a constraint-breaching (CB) class) *A non-rejectable Class $i \in \mathcal{I}$ having $\mathcal{L}_i = c_i/\theta_i$ is called a CB class, if $\tau_i < 1/\theta_i$. Let $\mathcal{I}^{CB} \subset \mathcal{I}$ denote the subset of all CB classes.*

In other words, a CB class is one where the average patience time is longer than the maximum allowable waiting time for this class, indicating that if all customers were allowed to wait according to their patience, the average waiting time would exceed the specified requirement. Our optimization problem can be written as follows:

$$\begin{aligned} &\max_{\bar{z},p,\hat{\theta}} \sum_{i=1}^{\mathcal{I}} \left[\hat{c}_i\mu_i\bar{z}_i + \lambda_i\left((\hat{c}_i - r_i)\,p_i - \hat{c}_i\right)\right] \\ &\text{s.t.} \quad \sum_{i\in\mathcal{I}} \bar{z}_i \leq 1, \\ &\qquad 0 \leq \bar{z}_i \leq \lambda_i\,(1-p_i)\,/\mu_i, \;\; i \in \mathcal{I} \\ &\qquad \lambda_i(1-p_i)\left(1-(\theta_i+\hat{\theta}_i)\tau_i\right)/\mu_i \leq \bar{z}_i, \;\; i \in \mathcal{I}^{CB}, \end{aligned} \tag{10}$$

where the last constraint is the SL requirement, which applies only to CB classes, since by definition it holds for all the other classes.

Lemma 1 (exclusive control usage for CB classes) *In the optimal solution of (10), for any specific set of τ_i, each CB class utilizes at most one control mechanism, either rejection or controlled abandonment, but not both.*

Next, we define the $\mathcal{L}^{\mathrm{S}}\mu$ rule, which will play a crucial role in our analysis:

Definition 4 (The $\mathcal{L}^{\mathrm{S}}\mu$ Rule) *Assign priority to Class i, $i \in \mathcal{I}$, having the higher $\mathcal{L}_i^S\mu_i$ index, where*

$$\mathcal{L}_i^S = \begin{cases} \mathcal{L}_i + 1_{\left\{\bar{z}_i \leq \frac{\lambda_i}{\mu_i}(1-\theta_i\tau_i)\right\}}\left(\hat{\alpha}_i - \mathcal{L}_i \wedge \frac{r_i-\mathcal{L}_i}{1-\tau_i\theta_i}\right), & \text{if } i \in \mathcal{I}^{CB}, \\ \mathcal{L}_i\,, & \text{Otherwise,} \end{cases}$$

Notice that $\left(\hat{\alpha}_i - \mathcal{L}_i \wedge \frac{r_i - \mathcal{L}_i}{1-\tau_i\theta_i}\right)$ is positive because for CB classes $\mathcal{L}_i = c_i/\theta_i$, meaning that $\mathcal{L}_i < \hat{\alpha}_i$, $\mathcal{L}_i < r_i$, and also $\tau_i < 1/\theta_i$.

According to the $\mathcal{L}^{\mathrm{S}}\mu$ rule, we allocate resources to a CB-Class i based on the adjusted index,

$$\mathcal{L}_i^{\mathrm{S}} = \mathcal{L}_i + \left(\hat{\alpha}_i - \mathcal{L}_i \wedge \frac{r_i - \mathcal{L}_i}{1-\tau_i\theta_i}\right),$$

until the allocated capacity reaches $\lambda_i(1-\theta_i\tau_i)/\mu_i$, which corresponds to a share of $(1-\theta_i\tau_i) \in (0,1]$ of the class's offered load λ_i/μ_i. Once this amount of capacity has been allocated, the SL constraint is satisfied, and the index for this class drops to $\mathcal{L}_i^{\mathrm{S}} = \mathcal{L}_i$, allowing for further resource allocation according to this updated index.

In case we still do not allocate enough capacity to satisfy the requirement, we have to either reject some of the customers or initiate a controlled abandonment. Choosing the minimum between $\hat{\alpha}_i - \mathcal{L}_i$ and $\frac{r_i - \mathcal{L}_i}{1-\tau_i\theta_i}$ determines the more cost-effective way to satisfy the SL requirement – controlled abandonment or rejection, respectively.

Theorem 2 establishes the optimality of the $\mathcal{L}^{\mathrm{S}}\mu$ index rule when incorporating SL requirements.

Theorem 2 (optimality under SL requirements) *For the optimization problem (10), which incorporates SL requirements, the $\mathcal{L}^S\mu$ rule is optimal. The adjusted optimal rejection probability is*

$$p_i = \begin{cases} 1_{\left\{\hat{\alpha}_i \geq \frac{r_i - \mathcal{L}_i\tau_i\theta_i}{1-\tau_i\theta_i}\right\}} \left(1 - \frac{\mu_i \bar{z}_i}{\lambda_i(1-\theta_i\tau_i)}\right), & \text{if } i \in \mathcal{I}^{CB}, \\ 1_{\left\{\frac{c_i}{\theta_i} \geq r_i,\ \hat{\alpha}_i \geq r_i\right\}} \left(1 - \frac{\mu_i}{\lambda_i}\bar{z}_i\right), & \text{otherwise.} \end{cases}$$

The adjusted optimal controlled abandonment rate is

$$\hat{\theta}_i = \begin{cases} 1_{\left\{\frac{r_i - \mathcal{L}_i\tau_i\theta_i}{1-\tau_i\theta_i} \geq \hat{\alpha}_i\right\}} \left(\frac{1}{\tau_i} - \frac{\mu_i \bar{z}_i}{\lambda_i\tau_i} - \theta_i\right), & \text{if } i \in \mathcal{I}^{CB}, \\ \infty, & \text{if } i \notin \mathcal{I}^{CB} \text{ and } \mathcal{L}_i = \hat{\alpha}_i, \\ 0, & \text{if } i \notin \mathcal{I}^{CB} \text{ and } \mathcal{L}_i \neq \hat{\alpha}_i. \end{cases}$$

5.1. Understanding the Mechanism of the $\mathcal{L}^{\mathrm{S}}\mu$ Rule

The key distinction between the $\mathcal{L}\mu$ and $\mathcal{L}^{\mathrm{S}}\mu$ rules is with regard to the CB classes. To ensure that these classes meet the SL constraints, the operator has three options: allocate additional resources, increase controlled abandonment rate, or increase rejection probability of arriving customers. Recall that according to Lemma 1 it is optimal to use at most one control. This trade-off is reflected in the elevated index for CB classes, which temporarily grants them higher priority when allocating resources. Once a class receives sufficient resources to satisfy the SL constraint, its index reverts to the standard $\mathcal{L}\mu$ value.

Figure 4 illustrates the fundamental difference between the $\mathcal{L}\mu$ and $\mathcal{L}^{\mathrm{S}}\mu$ rules within the fluid model. The total fluid capacity, represented by the large container in the figure, needs to be allocated among five customer classes, each depicted as a smaller container. The size of these containers reflects the capacity required by each class.

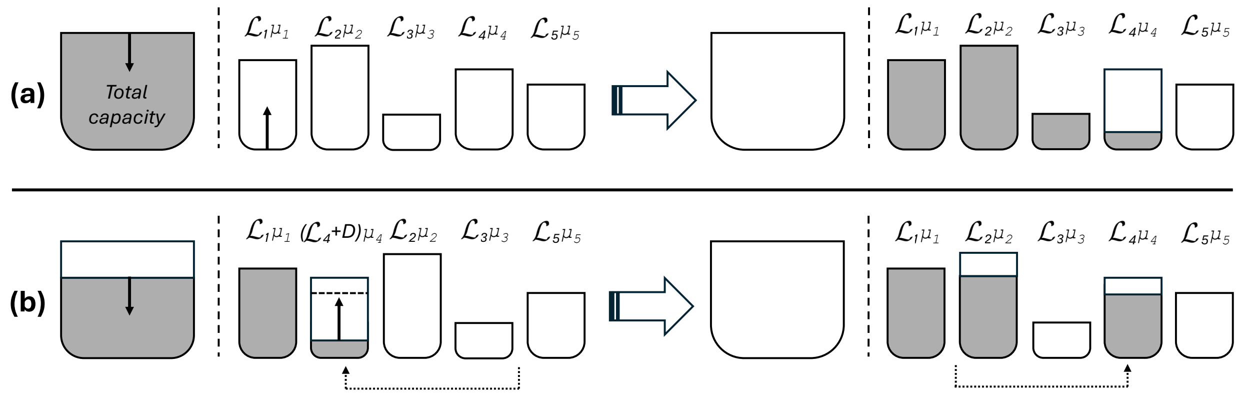

Figure 4 **Illustration of fluid allocation according to the $\mathcal{L}\mu$ rule (Case a) and the $\mathcal{L}^{\mathrm{S}}\mu$ rule (Case b).**

In Case **(a)**, the five classes are ordered according to their respective $\mathcal{L}\mu$ indices, with $\mathcal{L}_1\mu_1$ being the largest. The left side of the figure shows the initial stages of capacity allocation, while the right side shows the final allocation. Following the $\mathcal{L}\mu$ rule, capacity is allocated sequentially: first to Classes 1 through 3, with the remaining capacity going to Class 4.

Case **(b)** illustrates the allocation process under the $\mathcal{L}^{\mathrm{S}}\mu$ rule, where Class 4 is a CB class. According to the $\mathcal{L}^{\mathrm{S}}\mu$ rule, the index for Class 4 increases to $(\mathcal{L}_4 + D)\mu_4$, where $D := (r_4 - \mathcal{L}_4)/(1 - \tau_4\theta_4) \wedge \hat{\alpha}_4 - \mathcal{L}_4$, making it the second highest index after Class 1. Consequently, after allocating the necessary capacity to Class 1, capacity allocation shifts to Class 4 until the SL constraint is satisfied (indicated by the fluid reaching the dotted line within Class 4's container). At this point, Class 4's index reverts to $\mathcal{L}_4\mu_4$, and its priority drops, allowing the remaining capacity to be allocated to Class 2, leaving no capacity for Class 3.

5.2. Translation of the Fluid $\mathcal{L}^{\mathrm{S}}\mu$ Rule for the Stochastic System

The translation of the $\mathcal{L}^{\mathrm{S}}\mu$ rule is more complex than that of the $\mathcal{L}\mu$ rule due to the presence of CB classes and their index dependency on the allocated capacity $\bar{z}$. In general, customers with a higher $\mathcal{L}^{\mathrm{S}}\mu$ index are prioritized. For CB classes, this involves two distinct indices (see Definition 4). In the fluid model, capacity is allocated to CB classes by increasing the index until a $(1 - \theta_i\tau_i)$ share

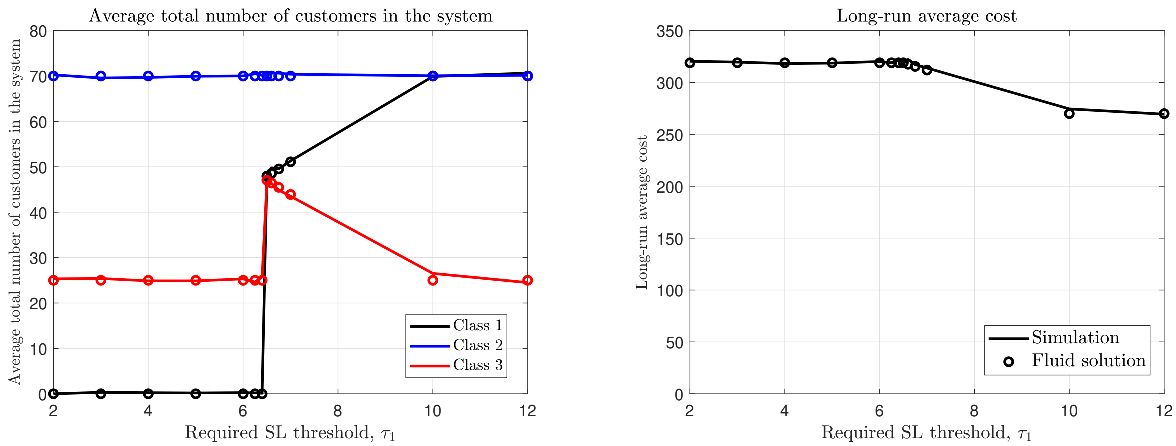

Figure 5 **Optimal number of customers in the system and optimal cost — Simulation of the $\mathcal{R}\mu$ rule versus the fluid solution (represented by circles) for varying service level (SL) thresholds. The parameters are:** $N = 5$, $\lambda = (7,7,7)$, $\mu = (1,1,1)$, $\theta = (0.1, 0.1, 0.1)$, $h = (0.8, 1.8, 2.8)$, $r = (17, 30, 30)$, $\alpha = (2,2,2)$, $\hat{\alpha} = (50, 50, 50)$.

of the class's requirements is served. The remaining $\theta_i\tau_i$ share of customers is served according to the original $\mathcal{L}\mu$ index. While there may be different interpretations of this policy in the stochastic system, we propose an intuitive one: predetermined capacity is allocated to a CB Class i so that the SL requirement is satisfied, i.e.,

$$\frac{\lambda_i}{\mu_i}(1-p_i)\left(1-(\theta_i+\hat{\theta}_i)\tau_i\right),$$

with the remaining capacity distributed among other classes according to their $\mathcal{L}\mu$ indices.

Figure 5 compares the performance of the $\mathcal{R}\mu$ rule, implemented in the stochastic system via simulation, with the fluid solution (represented by circles) across different levels of Class 1's SL threshold, τ_1.

When τ_1 is high, the majority of the system's capacity is allocated to Class 3. As τ_1 decreases, more capacity is shifted to Class 1 to meet its SL requirements, reducing the resources available for Class 3. However, when τ_1 becomes too low, Class 1 customers begin to be rejected, and capacity is reallocated to Class 3.

5.3. Impact of Service Level Requirements

We start this section by examining the effect of the SL required threshold on the optimal controlled abandonment rate and long-run average cost. Figure 6 provides two examples for a three-class setting, where Class 3 is a CB class (i.e., a non-rejectable class with $\mathcal{L}_3 = c_3/\theta_3$ and $\tau_3 < 1/\theta_3$). According to the basic $\mathcal{L}\mu$ indexes, Class 3 has the lowest priority among the three classes ($\mathcal{L}_1\mu_1 = 30$, $\mathcal{L}_2\mu_2 = 20$, $\mathcal{L}_3\mu_3 = 10$). However, once τ_3 drops below 10, the SL requirement for Class 3 is violated.

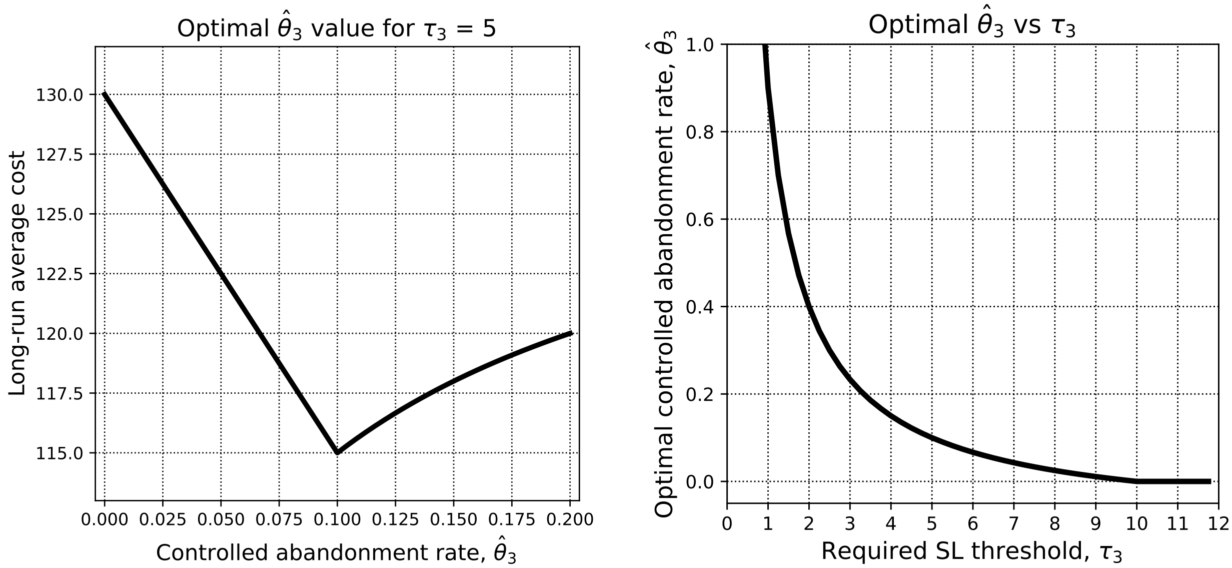

Figure 6 **An illustration of the relationship between the SL required threshold** τ_3**, the controlled abandonment rate** $\hat{\theta}$**, and the long-run average cost. The parameters are:** $N = 10$, $\lambda = (6, 6, 6)$, $\mu = (1, 1, 1)$, $\theta = (0.1, 0.1, 0.1)$, $h = (2.8, 1.8, 0.8)$, $r = (30, 30, 30)$, $\alpha = (2, 2, 2)$, $\hat{\alpha} = (50, 50, 15)$, $\tau = (10, 10, \tau_3)$**.**

Simply switching to the $\mathcal{L}^{\mathrm{S}}\mu$ indexes will not resolve the issue since, according to Definition 4, Class 3's index will increase to $\mathcal{L}^{\mathrm{S}}_3\mu_3 = \hat{\alpha}_3\mu_3 = 15$, but it will still remain last in prioritization. According to Theorem 2, meeting the SL requirement in this situation requires increasing the controlled abandonment rate $\hat{\theta}_3$.

Figure 6(a) shows the long-run average cost for different values of $\hat{\theta}_3$ when $\tau_3 = 5$. It is evident that $\hat{\theta}^*_3 = 0.1$ achieves the lowest cost, which is indeed the global optimum. In the SL requirement (9), the right-hand side becomes negative when $\hat{\theta}_i > 1/\tau_i - \theta_i$. Since $\bar{z}_i$ is non-negative, increasing $\hat{\theta}_i$ beyond this threshold negatively impacts the objective function.

Figure 6(b) shows the dependency between the optimal controlled abandonment rate and the threshold τ_3. Intuitively, as the threshold increases, it becomes easier to satisfy the SL requirement allowing the controlled abandonment rate to decrease. Note that when $\tau_3 \geq 10$, Class 3 is no longer a CB class, as the SL requirement is satisfied. The optimal rate in this range is, therefore, $\hat{\theta}^*_3 = 0$, which is aligned with the original $\mathcal{L}\mu$ rule.

Figure 7 presents three examples of the optimal server allocation (right plots) and long-run average cost (left plots) for different values of the SL threshold τ_3.

In Example 1, according to the $\mathcal{L}^{\mathrm{S}}\mu$ rule, Class 3 has the second-highest priority ($\mathcal{L}^{\mathrm{S}}_3 = \hat{\alpha}_3 = 28$). Thus, we identify three ranges with respect to τ_3:

i. When $\tau_3 \geq 10$: Class 3 is not a CB class, as the SL requirement is satisfied without allocating any resources to it. The solutions according to the $\mathcal{L}^{\mathrm{S}}\mu$ rule and the $\mathcal{L}\mu$ rule are identical.

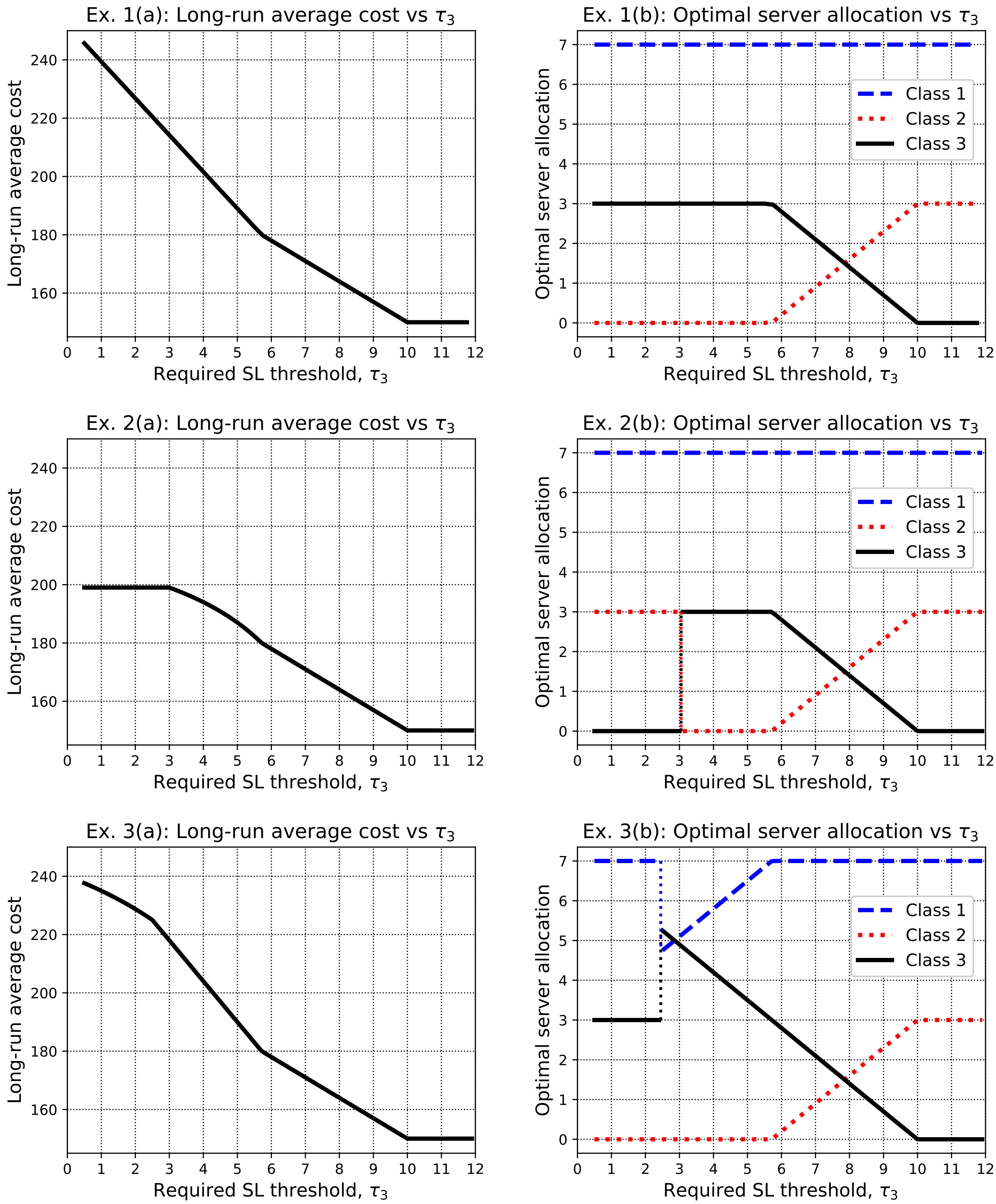

Figure 7 **Effect of** τ **on the cost and the optimal allocation.** $N = 10$, $\lambda = (7,7,7)$, $\mu = (1,1,1)$, $\theta = (0.1,0.1,0.1)$, $h = (2.8,1.8,0.8)$, $r = (30,30,r_3)$, $\alpha = (2,2,2)$, $\hat{\alpha} = (50,50,\hat{\alpha}_3)$, $\tau = (10,10,\tau_3)$**. Ex.** 1**: for controlled abandonment with** $r_3 = 30$, $\hat{\alpha}_3 = 28$**. Ex.** 2**: for rejection with** $r_3 = 17$, $\hat{\alpha}_3 = 50$**, Ex.** 3**: for rejection with** $r_3 = 25$, $\hat{\alpha}_3 = 50$**.**

ii. When $\tau_3 \in [40/7,\ 10)$: The SL requirement for Class 3 can be satisfied by allocating resources to Class 3 from Class 2, the next class in accordance with growing priority order. Once the constraint is satisfied, the remaining resources can be allocated to Class 2. In this range, as τ_3 decreases, more resources are allocated to Class 3 and fewer to Class 2.

iii. When $\tau_3 < 40/7$: There are no more resources for Class 2 that we can allocate to Class 3 to satisfy the SL requirement (see (9)). To address this, the controlled abandonment rate $\hat{\theta}_3$ must be increased as well, which sharply increases the cost. Therefore, in this range, the allocation is fixed — all resources from Class 2 are allocated to Class 3. However, as τ_3 decreases, $\hat{\theta}_3$ increases.

In Example 2, we increase $\hat{\alpha}_3$ to 50 so that controlled abandonment is knowingly not optimal, and take $r_3 = 17$. Cases (i) and (ii) remain the same as in Example 1, but case (iii) is now divided into two:

iii.a. When $\tau_3 \in [3,\ 40/7)$: This time to keep satisfying the SL requirement after Class 3 acquired all the capacity from Class 2, we start rejecting Class 3 patients. Note, that the cost here grows non-linearly due to the Definition 4, where the additional cost of rejection reduces as τ gets smaller.

iii.b. When $\tau_3 < 3$: The additional cost of rejection reduces to the point where it is cheaper to reject than to reallocate capacity from Class 2, leading to instant complete rejection of Class 3 and thus no additional cost with further reduction of τ_3.

In Example 3 we maintain knowingly high $\hat{\alpha}_3$, while also increase rejection cost to $r_3 = 25$. Again, cases (i) and (ii) remain the same, while case (iii) is divided into two:

iii.a. When $\tau_3 \in [5/2,\ 40/7)$: This time the optimal way to keep satisfying the SL requirement after Class 3 acquired all the capacity from Class 2, is to start reallocating capacity to Class 3 from Class 1. This leads to a sharper linear growth in optimal cost.

iii.b. When $\tau_3 < 5/2$: The additional cost of rejection reduces to the point where it's cheaper to reject than to reallocate capacity from Class 1, but not from Class 2. This leads to an instant release of Class 1 capacity and familiar non-linear growth of cost due to rejection.

Note that the optimal server allocation and the respective optimal long-run cost are the same for $\tau \geq 40/7$ in all the examples. At the same time they allow us to compare how different values of $\hat{\alpha}_3$ and r_3 affect the performances when $\tau_3 < 40/7$.

6. Concluding Remarks and Future Research Directions

We study the optimal scheduling policy for multi-server queues with multiple customer classes, focusing on two key operational controls: customer rejection upon arrival and controlled abandonment during waiting times. These controls enable the operator to minimize costs effectively.

Our model is broadly applicable across various domains, including service systems, call centers, healthcare, and cloud services.

By developing a corresponding fluid model, we derive the $\mathcal{L}\mu$ rule – an index-based policy that efficiently balances holding costs, service rates, and both customer-initiated and operator-initiated abandonment rates. Under optimal capacity allocation, each customer class operates either as an Erlang-B queue, where customers are rejected if immediate service is unavailable, or as an Erlang-A queue, where customers wait for service or abandon the queue.

In addition to being asymptotically optimal in the heavy-traffic, many-server regime, simulation experiments demonstrate the effectiveness of the $\mathcal{L}\mu$ rule in systems of different size.

When class-specific service level (SL) requirements are introduced, the $\mathcal{L}\mu$ rule is adapted into the $\mathcal{L}^{\mathrm{S}}\mu$ rule, which incorporates SL constraints into the prioritization process. Under the $\mathcal{L}^{\mathrm{S}}\mu$ rule, strict priority is relaxed, allowing for the possibility that a class may not receive all the capacity it demands. In such cases, part of capacity is allocated to a class with a lower $\mathcal{L}\mu$ index but a stricter SL requirement. Translating the fluid-based policy back into the original stochastic system results in a two-stage index policy, ensuring cost-effective scheduling while adhering to all SL constraints.

Looking ahead, there are several promising directions for future research. One potential avenue involves developing policies for time-varying systems, where arrival rates and staffing levels vary over time. The idea would be to incorporate these dynamics into scheduling decisions, along with time-varying rejection and controlled abandonment strategies. Another direction could explore extensions to more complex network settings, such as tandem networks, where customers may be rejected at different stages, or parallel-station networks, where customers might be redirected to alternative stations instead of being rejected. Finally, a promising direction is to develop a decision-making tool for defining the best allowed maximal average waiting time (τ) parameter. For this we could incorporate the benefit of utilizing SL requirement into the cost function, which would allow us to investigate the trade-off between reducing τ and the increase in operational cost it implies.

Declarations. Conflict of interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A: Proofs of Analytical Results

Proof of Proposition 1: We start with the proof of $\inf_{\hat{\theta}_i} \hat{c}_i$ value. Let us take the following derivative:

$$(\hat{c}_i)'_{\hat{\theta}_i} = \frac{\hat{\alpha}_i(\theta_i + \hat{\theta}_i) - (c_i + \hat{\alpha}_i\hat{\theta}_i)}{\left(\theta_i + \hat{\theta}_i\right)^2} = \frac{\hat{\alpha}_i\theta_i - c_i}{\left(\theta_i + \hat{\theta}_i\right)^2}.$$

Since we are interested in comparing the derivative to zero, and the denominator is positive because we assume $\theta_i > 0$, we only need to consider the numerator. The numerator, which is independent of $\hat{\theta}_i$, is either non-negative or non-positive. With $0 \le \hat{\theta}_i < \infty$, we find that $\hat{c}_i$ always lies between $\hat{\alpha}_i$ (when $\hat{\theta}_i \to \infty$) and c_i/θ_i (when $\hat{\theta}_i = 0$). We consider the following cases:

1. $\hat{\alpha}_i < c_i/\theta_i \;\Rightarrow\; (\hat{c}_i)'_{\hat{\theta}_i} < 0 \;\Rightarrow\; \hat{c}_i$ is a decreasing function $\Rightarrow$ $\inf_{\hat{\theta}_i} \hat{c}_i \; = \hat{\alpha}_i$, when $\hat{\theta}_i \to \infty$.
2. $\hat{\alpha}_i = c_i/\theta_i \;\Rightarrow\; (\hat{c}_i)'_{\hat{\theta}_i} = 0 \;\Rightarrow\; \hat{c}_i$ is a constant function $\Rightarrow$ $\inf_{\hat{\theta}_i} \hat{c}_i = \hat{\alpha}_i = c_i/\theta_i, \; \forall \hat{\theta}_i \ge 0$.
3. $\hat{\alpha}_i > c_i/\theta_i \;\Rightarrow\; (\hat{c}_i)'_{\hat{\theta}_i} > 0 \;\Rightarrow\; \hat{c}_i$ is an increasing function $\Rightarrow$ $\inf_{\hat{\theta}_i} \hat{c}_i \; = c_i/\theta_i$, when $\hat{\theta}_i = 0$,

as stated.

Proving $\hat{c}_i^* = \inf_{\hat{\theta}_i} \hat{c}_i$ is based on (4). The first line can be rewritten as

$$\max_{\bar{z},p,\hat{\theta}} \sum_{i=1}^{\mathcal{I}} \left[\hat{c}_i \left(\mu_i \bar{z}_i - \lambda_i(1-p_i)\right) - \lambda_i r_i p_i\right],$$

where $\mu_i \bar{z}_i - \lambda_i(1-p_i) \le 0$ by the problem definition and thus for any fixed p_i we get that $\hat{c}_i = \inf_{\hat{\theta}_i} \hat{c}_i$ is optimal.

Next, we turn to the rejectable classes.

The proof follows from (4) by demonstrating that for any solution where $\bar{q}_i = a$, for some $a > 0$, a modified solution with $\bar{q}_i = 0$ and an increased p_i (according to (2)) achieves a better value function.

For $\bar{q}_i = a > 0$ and $p_i = p^a$, we have

$$\lambda_i = \mu_i \bar{z}_i + \left(\theta_i + \hat{\theta}_i\right) a + \lambda_i p_i^a,$$

with the following value function:

$$F^a = \left(c_i + \hat{\alpha}_i \hat{\theta}_i\right) a + r_i \lambda_i p_i^a + \sum_{j \ne i}^{\mathcal{I}} \left[c_j \bar{q}_j + \hat{\alpha}_j \hat{\theta}_j \bar{q}_j + r_j \lambda_j p_j\right]$$

For $\bar{q}_i = 0$ and $p_i = p_i^0$, we have

$$\lambda_i = \mu_i \bar{z}_i + \lambda_i p_i^0,$$

with the following value function:

$$F^0 = r_i \lambda_i p_i^0 + \sum_{j \ne i}^{\mathcal{I}} \left[c_j \bar{q}_j + \hat{\alpha}_j \hat{\theta}_j \bar{q}_j + r_j \lambda_j p_j\right].$$

Equating expressions for λ_i, we get

$$p_i^0 = p_i^a + \frac{a}{\lambda_i}(\theta_i + \hat{\theta}_i).$$

Then, comparing the two value functions, we have

$$\begin{aligned}
F^a - F^0 &= \left(c_i + \hat{\alpha}_i \hat{\theta}_i\right) a + r_i \lambda_i p_i^a - r_i \lambda_i p_i^0 \\
&= \left(c_i + \hat{\alpha}_i \hat{\theta}_i\right) a + r_i \lambda_i p_i^a - r_i \lambda_i p_i^a - r_i a \left(\theta_i + \hat{\theta}_i\right) \\
&= \left(c_i + \hat{\alpha}_i \hat{\theta}_i\right) a - r_i a \left(\theta_i + \hat{\theta}_i\right) \\
&= a \left(\left(c_i + \hat{\alpha}_i \hat{\theta}_i\right) - r_i \left(\theta_i + \hat{\theta}_i\right)\right) \\
&\geq a \left(\left(c_i + \hat{\alpha}_i \hat{\theta}_i\right) - \hat{c}_i \left(\theta_i + \hat{\theta}_i\right)\right) = 0,
\end{aligned}$$

where the first inequality is derived from proposition's condition that $r_i \leq \hat{c}_i$, and the second is from the definition of $\hat{c}_i$. This proves that when $r_i \leq \hat{c}_i$, $\bar{q}_i = 0$ is preferable.

Note that in this case we can plug in $\bar{q}_i = 0$ into (2) and get that $\bar{z}_i = \frac{\lambda_i}{\mu_i}(1 - p_i)$, which is equivalent to $p_i = 1 - \frac{\mu_i}{\lambda_i} \bar{z}_i$.

Lastly, we move to the non-rejectable classes. Since $\hat{c}_i^* < r_i$ then $p_i^* = 0$ follows from (4).

Q.E.D.

Proof of Theorem 1: Let us separately consider the rejectable classes $L \subset \mathcal{I}$ and the non-rejectable classes $K \subset \mathcal{I}$. Note that $L \cup K = \mathcal{I}$ and $L \cap K = \varnothing$. For each rejectable class $l \in L$ (i.e., $\mathcal{L}_l = r_l$), we substitute $p_l = 1 - \frac{\mu_l \bar{z}_l}{\lambda_l}$ (as derived in Proposition 1) into the objective function of (4). For each non-rejectable class $k \in K$ (i.e., $\mathcal{L}_k = \hat{\alpha}_k$ or $\mathcal{L}_k = \frac{c_k}{\theta_k}$), we substitute $p_k = 0$ (as derived in Proposition 1). We, therefore, have:

$$\begin{aligned}
&\min_{\bar{z}, p, \hat{\theta}} \left[\sum_{l \in L} (\lambda_l r_l - r_l \mu_l \bar{z}_l) + \sum_{k \in K} \left(\lambda_k \inf_{\hat{\theta}_k} \hat{c}_k - \inf_{\hat{\theta}_k} \hat{c}_k \mu_k \bar{z}_k\right)\right] \Leftrightarrow \quad \max_{\bar{z}} \left[\sum_{l \in L} \mathcal{L}_l \mu_l \bar{z}_l + \sum_{k \in K} \mathcal{L}_k \mu_k \bar{z}_k\right] \\
&\text{s.t.} \quad \sum_{i \in \mathcal{I}} \bar{z}_i \leq 1, \quad \lambda_i / \mu_i \geq \bar{z}_i \geq 0,
\end{aligned}$$

where equivalence is due to constant elimination and $\mathcal{L}$ definition. Clearly, the optimal solution for the above problem is to allocate the servers according to their $\mathcal{L}\mu$ index. Q.E.D.

Proof of Proposition 2. We start by defining the fluid scaled processes:

$$\begin{aligned}
&\bar{X}^n = \frac{1}{n} X^n, \ \bar{Q}^n = \frac{1}{n} Q^n, \ \bar{Z}^n = \frac{1}{n} Z^n, \\
&\bar{A}^n = \frac{1}{n} A^n, \ \bar{A}^{(P)n} = \frac{1}{n} A^{(P)n}, \ \bar{A}^{(1-P)n} = \frac{1}{n} A^{(1-P)n}, \\
&\bar{R}^n = \frac{1}{n} R^n, \ \bar{D}^n = \frac{1}{n} D^n,
\end{aligned}$$

which satisfy the following

$$\begin{aligned}
&\sum_{i \in \mathcal{I}} \bar{Z}_i^n(t) \leq 1 \\
&\bar{Q}_i^n(t) = \bar{X}_i^n(t) - \bar{Z}_i^n(t) \\
&\bar{X}_i^n(t) = \bar{X}_i^n(0) + \bar{A}_i^{(1-P)n}(t) - \bar{R}_i^n(t) - \bar{D}_i^n(t).
\end{aligned}$$

Given T and $\delta \in (0,1)$, we define the event $E^n = E^n_{\delta,T}$ as

$$E^n = E^n_{A(1-P)} \cap E^n_{A(P)} \cap E^n_D \cap E^n_R,$$

where

$$E^n_{A(1-P)} = \left\{ \max_i \sup_{t\in[0,T]} \left| \bar{A}^{(1-P)n}_i(t) - \frac{(1-P^n_i)\lambda^n_i}{n} t \right| < \delta \right\},$$

$$E^n_{A(P)} = \left\{ \max_i \sup_{t\in[0,T]} \left| \bar{A}^{(P)n}_i(t) - \frac{P^n_i\lambda^n_i}{n} t \right| < \delta \right\},$$

$$E^n_D = \left\{ \max_i \sup_{t\in[0,T]} \left| \frac{\tilde{D}^n_i(nt)}{n} - \mu^n_i t \right| < \delta \right\},$$

$$E^n_R = \left\{ \max_i \sup_{t\in[0,KT]} \left| \frac{\tilde{R}^n_i(nt)}{n} - \theta^n_i t \right| < \delta \right\}.$$

Here, $K = K(T) = \frac{1}{2}c_\lambda T + M + 1$, where $c_\lambda = \sup_n \frac{\lambda_n}{n} < \infty$, and $M < \infty$ is a bound on $n^{-1}\|X^n(0)\|$. From this point on we use notation $\bar{\lambda}^n_i := \lambda^n_i/n$.

The proof follows a similar line of arguments as in Atar et al. 2010, Propositions 4.1 and Theorem 5.1, with the following adaptations:

(1) In our case, the cumulative number of arrivals, A^n_i, is divided into those who were rejected, $A^{(P)n}_i$, and those who entered the system, $A^{(1-P)n}_i$. Therefore, throughout the proof, we replace the cumulative number of arrivals, A^n_i, with the effective cumulative arrivals $A^{(1-P)n}_i$, as in (7). Hence, we use $\bar{A}^{(1-P)n}_i$, $(1-P^n_i)\lambda^n_i$, and $(1-P^n_i)\bar{\lambda}^n_i$ instead of $\bar{A}^n_i$, λ^n_i, and $\bar{\lambda}^n_i$, respectively. Given that by definition $A^{(1-P)n}_i \leq A^n_i$ and $(1-P^n_i)\lambda^n_i \leq \lambda^n_i$, all corresponding equations and inequalities are preserved.
Furthermore, one of the claims stated in the proof is that $\mathbb{P}\left(E^n_{A(1-P)}\right) \to 1$ and $\mathbb{P}\left(E^n_{A(P)}\right) \to 1$ as $n \to \infty$ (instead of $\mathbb{P}(E^n_A) \to 1$ in the proof of Atar et al. 2010, Lemma 2.1). This is established by noting that $A^{(P)n}_i$ and $A^{(1-P)n}_i$ are also Poisson processes with rates $\lambda^n_i P^n_i$ and $\lambda^n_i(1-P^n_i)$, respectively. Since $P^n_i \to p_i$ and $\lambda^n_i/n \to \lambda_i$ as $n \to \infty$ by Assumption 1, it follows that

$$\lim_{n\to\infty} \frac{\lambda^n_i P^n_i}{n} = \lambda_i p_i \quad \text{and} \quad \lim_{n\to\infty} \frac{\lambda^n_i(1-P^n_i)}{n} = \lambda_i(1-p_i).$$

(2) In our case, both the stochastic and fluid objective functions incorporate an additional term that accounts for the cost of rejection. Specifically, in the normalized average cost in (8), the following term represents the average rejection cost:

$$\frac{1}{nT}\mathbb{E}\left[rA^{(P)n}(T)\right],$$

which in the fluid objective function corresponds to $r\lambda p$.
In what follows, we provide the necessary adaptations to the proof for both part (a) and part (b) to fit our case.

Part (a). The proof of the first part of Proposition 2 follows the same line of arguments used in Atar et al. 2010, Proposition 4.1, with necessary adaptations. The proof concludes by establishing that for any fixed $\epsilon > 0$, δ and a sufficiently large T, there exists an $n_0 = n_0(\epsilon)$, where for all $n \geq n_0$, on the event $E^n = E^n_{\delta,T}$, the inequality $cq^n \geq cq^* - \rho(\epsilon)$ holds, where $\rho : [0,\infty) \to [0,\infty)$ is a continuous function that vanishes at zero.

In our case, we have:

$$V^* = cq^* + r\lambda p^*,$$

where the second term, which we focus on, represents the average fluid rejection cost.

On the event E^n, we have $\forall i \in \mathcal{I}:\ \bar{A}_i^{(P)n}(T)/T \to \bar{\lambda}_i^n P_i^n$ as $T \to \infty$. By Assumption 1, $\bar{\lambda}_i^n P_i^n \to \lambda_i p_i$ as $n \to \infty$. Within the framework of the original proof, it follows that there exists a sufficiently large T and $n_0 = n_0(\epsilon)$ such that for all $n \geq n_0$, on the event E^n, we have:

$$cq^n + \frac{r}{T}\bar{A}_i^{(P)n}(T) \geq cq^* + r\lambda p - \rho(\epsilon) = V^* - \rho(\epsilon).$$

Thus,

$$C_{n,T}(\pi^n) = \mathbb{E}\left[cq^n + \frac{r}{T}\bar{A}_i^{(P)n}(T)\right] \geq \mathbb{E}\left[1_{E^n}\left(cq^n + \frac{r}{T}\bar{A}_i^{(P)n}(T)\right)\right],$$

and

$$\liminf_n C_{n,T}(\pi^n) \geq (V^* - \rho(\epsilon)) \liminf_n \mathbb{P}(E^n) = V^* - \rho(\epsilon),$$

which completes the proof of this part.

Part (b): The proof is based on Atar et al. 2010, Theorem 5.1, with a few necessary adjustments.

Lemma 5.1 in that paper asserts that for every $\epsilon > 0$, there exists $T_\epsilon \in (0, \infty)$ such that for all $T \in (T_\epsilon, \infty)$,

$$\sup_{t \in [T_\epsilon, T]} \left\|\bar{Z}^n(t) - z^*\right\| \vee \left\|\bar{Q}^n(t) - q^*\right\| < \epsilon$$

holds on the event $E_{\delta,T}^n$ for all $n \geq n_0$, where both δ and n_0 depend on ϵ and T.

Returning to our proof, we fix $\epsilon > 0$, $T > T_\epsilon$, and retain the definitions of δ and n_0.

Recall that

$$C_{n,T}(\pi^n) = \frac{1}{T}\mathbb{E}\left[\int_0^T c\ \bar{Q}^n(t)\,dt + r\bar{A}^{(P)n}(T)\right].$$

We then have

$$\begin{aligned} C_{n,T}(\pi_{\mathcal{L}}^n) \leq & \frac{1}{T}\mathbb{E}\left[1_{E_{\delta,T}^n}\int_0^T c\ \bar{Q}^n(t)\,dt\right] + \frac{1}{T}\mathbb{E}\left[1_{E_{\delta,T}^n} r\bar{A}^{(P)n}(T)\right] \\ & + \mathbb{E}\left[1_{(E_{\delta,T}^n)^c}\|c\ \bar{Q}^n\|_T^*\right] + \mathbb{E}\left[1_{(E_{\delta,T}^n)^c}\|r\bar{A}^{(P)n}(T)\|_T^*\right], \end{aligned} \tag{11}$$

where $1_{(E_{\delta,T}^n)^c}$ denotes the indicator function for the complement of the event $E_{\delta,T}^n$, and for $f : \mathbb{R}_+ \to \mathbb{R}^{\mathcal{I}}$, we define $\|f\|_T^* = \sup_{0 \leq t \leq T}\|f(t)\|$.

Taking the limits $n \to \infty$ followed by $T \to \infty$, we obtain

$$v_p := \limsup_{T\to\infty}\limsup_{n\to\infty} C_{n,T}(\pi_{\mathcal{L}\mu}^n) \leq cq^* + r\lambda p + \|c\|\epsilon + \|r\|\epsilon = V^* + (\|c\| + \|r\|)\epsilon,$$

where the inequality holds since the last two terms in (11) diminish. For the second term, we use the fact that $|\bar{A}_i^{(P)n}(T)/T - \lambda p| < \epsilon$, and the first term is the same as in Atar et al. 2010, Theorem 5.1. Since $\epsilon > 0$ is arbitrary, part (b) is proven, and the proof is complete. Q.E.D.

Proof of Lemma 1: Let us consider the class that does not satisfy SL requirement given original $\mathcal{L}\mu$ index for the class. In order to satisfy the requirement we consider the trade-off between reallocating capacity from other classes and using our controls to reduce the amount of clients waiting in the queue. Satisfying the constraint for such class means that $\hat{z} = \lambda(1-p)\left(1 - (\theta + \hat{\theta})\tau\right)/\mu$. Let us take derivatives:

$$\hat{z}'_p = -\frac{\lambda}{\mu}\left(1 - (\theta + \hat{\theta})\tau\right); \quad \hat{z}'_{\hat{\theta}} = -\frac{\lambda}{\mu}(1-p)\tau,$$

where $\hat{z}'_p \leq 0$ because $\hat{z} \geq 0$ meaning that $1-(\theta+\hat{\theta})\tau \geq 0$.

Both derivatives show that the effectiveness of one control is getting smaller when the other control is used more, meaning that if one of the controls is more effective (cost-wise) than the other when both are zeros, it will always be more effective. Q.E.D.

Proof of Theorem 2: The idea of the proof is similar to that of Theorem 1. However, now we must consider the CB classes separately. Let $L \subset \mathcal{I}$ denote all rejectable classes (i.e., such $l \in \mathcal{I}$ that $\mathcal{L}_l = r_l$) and $K^{SL} \subset \mathcal{I}$ denote those non-rejectable classes $k \in \mathcal{I}$ that have either $\mathcal{L}_k = c_k/\theta_k$ with $\tau_k \geq 1/\theta_k$ or $\mathcal{L}_k = \hat{\alpha}_k$. Additionally, we consider the CB classes $\mathcal{I}^{CB} \subset \mathcal{I}$ (i.e., such that for all $m \in \mathcal{I}^{CB}$, $\mathcal{L}_m = c_m/\theta_m$ and $\tau_m < 1/\theta_m$). Note that

$$L \cup K^{SL} \cup \mathcal{I}^{CB} = \mathcal{I} \quad \text{ and } \quad L \cap K^{SL} = K^{SL} \cap \mathcal{I}^{CB} = \mathcal{I}^{CB} \cap L = \varnothing.$$

For each class $l \in L$ we insert $p_l = 1 - \mu_l \bar{z}_l/\lambda_l$ into the objective function of (10). For each class $k \in K^{SL}$ we insert $p_k = 0$:

$$\begin{aligned}
&\max_{\bar{z},p,\hat{\theta}} \left[\sum_{l \in L} (r_l \mu_l \bar{z}_l - \lambda_l r_l) + \sum_{k \in K^{SL}} (\mathcal{L}_k \mu_k \bar{z}_k - \lambda_k \mathcal{L}_k) + \sum_{m \in \mathcal{I}^{CB}} (\hat{c}_m \mu_m \bar{z}_m + \lambda_m (\hat{c}_m - r_m) p_m - \lambda_m \hat{c}_m) \right] \\
\Leftrightarrow \quad &\max_{\bar{z},p} \left[\sum_{l \in L} r_l \mu_l \bar{z}_l + \sum_{k \in K^{SL}} \mathcal{L}_k \mu_k \bar{z}_k + \sum_{m \in \mathcal{I}^{CB}} (\hat{c}_m \mu_m \bar{z}_m - \lambda_m (r_m - \hat{c}_m) p_m - \lambda_m \hat{c}_m) \right] \\
\text{s.t.} \quad & \sum_{i \in \mathcal{I}} \bar{z}_i \leq 1, \quad \frac{\lambda_i}{\mu_i} \geq \bar{z}_i \geq 0, \ i \in \mathcal{I} \\
& \bar{z}_m > \lambda_m (1-p_m)(1-(\theta_m + \hat{\theta}_m)\tau_m)/\mu_m \text{ for } m \in \mathcal{I}^{CB};
\end{aligned}$$

Let us look closer at the CB classes. According to Lemma 1 in optimal solution for each CB class m_0 we have two cases: use only rejection control ($\hat{\theta}_{m_0} = 0$) and use only controlled abandonment ($p_{m_0} = 0$). We denote $M := \sum_{m \in \mathcal{I}^{CB} | m \neq m_0} (\hat{c}_m \mu_m \bar{z}_m - \lambda_m (r_m - \hat{c}_m) p_m - \lambda_m \hat{c}_m)$

Rejection case: $\hat{\theta}_{m_0} = 0$. We note that for any fixed value of $\bar{z}_{m_0}$, minimizing p_{m_0}, results in two cases:

(a) $\boldsymbol{\bar{z}_{m_0} < (1-\theta_{m_0}\tau_{m_0})\lambda_{m_0}/\mu_{m_0}}$: The optimal solution is

$$p_{m_0} = 1 - \frac{\mu_{m_0} \bar{z}_{m_0}}{\lambda_{m_0}(1-\theta_{m_0}\tau_{m_0})}$$

Substituting this into our objective function will yield the following:

$$\max_{\bar{z}} \left[\sum_{l \in L} r_l \mu_l \bar{z}_l + \sum_{k \in K^{SL}} \mathcal{L}_k \mu_k \bar{z}_k + \left(\frac{c_{m_0}}{\theta_{m_0}} \mu_{m_0} \bar{z}_{m_0} - \lambda_{m_0} r_{m_0} + \frac{r_{m_0} - \frac{c_{m_0}}{\theta_{m_0}}}{1-\theta_{m_0}\tau_{m_0}} \mu_{m_0} \bar{z}_{m_0} \right) + M \right],$$

where the middle addend for the CB classes is a constant and does not affect the solution. We, therefore, get that

$$\max_{\bar{z}} \left[\sum_{l \in L} \mathcal{L}_l \mu_l \bar{z}_l + \sum_{k \in K^{SL}} \mathcal{L}_k \mu_k \bar{z}_k + \left(\mathcal{L}_{m_0} + \frac{r_{m_0} - \mathcal{L}_{m_0}}{1-\theta_{m_0}\tau_{m_0}} \right) \mu_{m_0} \bar{z}_{m_0} + M \right].$$

(b) $\boldsymbol{\bar{z}_{m_0} \geq (1-\theta_{m_0}\tau_{m_0})\lambda_{m_0}/\mu_{m_0}}$: The optimal solution would be $p_{m_0} = 0$, so the objective function will be:

$$\max_{\bar{z}} \left[\sum_{l \in L} \mathcal{L}_l \mu_l \bar{z}_l + \sum_{k \in K^{SL}} \mathcal{L}_k \mu_k \bar{z}_k + \mathcal{L}_{m_0} \mu_{m_0} \bar{z}_{m_0} + M \right],$$

Controlled abandonment case: $p_{m_0} = 0$. We note that for any fixed value of $\bar{z}_{m_0}$, we want to minimize $\hat{c}_{m_0}$ due to $\bar{z}_{m_0} \leq \lambda_{m_0}/\mu_{m_0}$, meaning minimizing $\hat{\theta}_{m_0}$. This again results in two similar cases:

(a) $\boldsymbol{\bar{z}_{m_0} < (1 - \theta_{m_0}\tau_{m_0})\lambda_{m_0}/\mu_{m_0}}$: The optimal solution is

$$\hat{\theta}_{m_0} = \frac{1}{\tau_{m_0}} - \frac{\mu_{m_0}\bar{z}_{m_0}}{\lambda_{m_0}\tau_{m_0}} - \theta_{m_0}$$

Substituting this into our objective function will yield the following:

$$\max_{\bar{z}} \left[\sum_{l \in L} \mathcal{L}_l \mu_l \bar{z}_l + \sum_{k \in K^{SL}} \mathcal{L}_k \mu_k \bar{z}_k + \left(\frac{c_{m_0} + \frac{\hat{\alpha}_{m_0}}{\tau_{m_0}} \left(1 - \frac{\mu_{m_0}}{\lambda_{m_0}} \bar{z}_{m_0}\right) - \hat{\alpha}_{m_0}\theta_{m_0}}{\frac{1}{\tau_{m_0}} \left(1 - \frac{\mu_{m_0}}{\lambda_{m_0}} \bar{z}_{m_0}\right)} \left(1 - \frac{\mu_{m_0}}{\lambda_{m_0}} \bar{z}_{m_0}\right) (-\lambda_{m_0}) \right) + M \right],$$

from where we get

$$\max_{\bar{z}} \left[\sum_{l \in L} \mathcal{L}_l \mu_l \bar{z}_l + \sum_{k \in K^{SL}} \mathcal{L}_k \mu_k \bar{z}_k + \left(c_{m_0}\tau_{m_0} + \hat{\alpha}_{m_0} \left(1 - \frac{\mu_{m_0}}{\lambda_{m_0}} \bar{z}_{m_0}\right) - \hat{\alpha}_{m_0}\theta_{m_0}\tau_{m_0} \right) (-\lambda_{m_0}) + M \right],$$

and then, by eliminating constants, to an equivalent problem

$$\max_{\bar{z}} \left[\sum_{l \in L} \mathcal{L}_l \mu_l \bar{z}_l + \sum_{k \in K^{SL}} \mathcal{L}_k \mu_k \bar{z}_k + \hat{\alpha}_{m_0}\mu_{m_0}\bar{z}_{m_0} + M \right].$$

(b) $\boldsymbol{\bar{z}_{m_0} \geq (1 - \theta_{m_0}\tau_{m_0})\lambda_{m_0}/\mu_{m_0}}$: The optimal solution would be $\hat{\theta}_{m_0} = 0$, so the objective function will be:

$$\max_{\bar{z}} \left[\sum_{l \in L} \mathcal{L}_l \mu_l \bar{z}_l + \sum_{k \in K^{SL}} \mathcal{L}_k \mu_k \bar{z}_k + \mathcal{L}_{m_0}\mu_{m_0}\bar{z}_{m_0} + M \right],$$

Those cases mean that if the allocated capacity for a CB class m is not enough to satisfy the SL constraint, we will choose the modified $\mathcal{L}_m^{\mathrm{S}}\mu_m$ index for that class, where $\mathcal{L}_m^{\mathrm{S}}$ is the minimum between $\hat{\alpha}_m = \mathcal{L}_m + (\hat{\alpha}_m - \mathcal{L}_m)$, corresponding to controlled abandonment, and $\mathcal{L}_m + \frac{r_m - \mathcal{L}_m}{1 - \tau_m\theta_m}$, corresponding to rejection. Q.E.D.

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